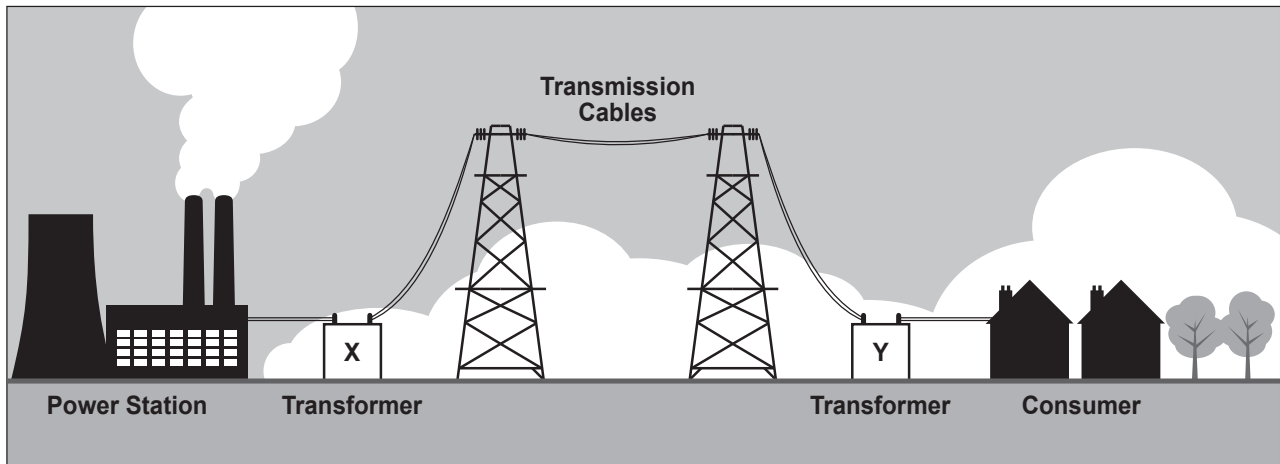


4. The diagram below shows part of the National Grid.



- (a) Complete the sentences about the National Grid by selecting the correct words from the box. Each word can be used once, more than once or not at all. [4]

faster	reliable	increases	generates	safely	decreases
--------	----------	-----------	-----------	--------	-----------

The National Grid makes sure that the electricity supply to consumers is

.....

Transformer X the voltage and
the current.

Transformer Y changes the voltage so that consumers can use it
..... in their homes.



- (b) Five electrical supply companies (**A**, **B**, **C**, **D** and **E**) each produce the same amount of electrical power.

The table below shows the percentage of electrical power output from each energy source by each company.

Percentage of electrical power output from each energy source (%)					
Energy source	Company A	Company B	Company C	Company D	Company E
coal	4.8	5.8	7.1	10.8	10.1
natural gas	63.0	55.5	9.6	63.0	46.0
nuclear	5.2	20.6	71.2	23.3	13.9
renewable	27.0	18.1	12.1	2.9	30.0

Use the table to answer the following questions.

- (i) State which energy company produces the most electrical power from coal. [1]

.....

- (ii) State which company produces the most nuclear waste. [1]

.....

- (iii) Calculate the **mean** percentage of electrical power output generated from **coal** by the five companies (**A**, **B**, **C**, **D** and **E**). [2]

mean percentage =

- (c) The carbon footprint of an energy supply customer can be an important factor when choosing a supply company.

- (i) State what is meant by the term 'carbon footprint'. [2]

.....

.....

- (ii) A customer decides against using Company **A** since they believe that they have the highest carbon footprint from generating electricity.

Use information from the table to explain whether the customer is correct. [2]

.....

.....

.....

